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" On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues."



Q1 => The state transition matrix for a dicrete system is given as 'A'. Show that the system is not stable if the determinant value of the matrix 'A' is greater than one (10p)

Q2 Find a non-zero function which is orthogonal to the following functions $(0,1)$; x^2, x^4. How many different functions satisfying the requirement can be found? Explain

Q3 Consider the system given in the figure below, here tank A is full of water and tank B is empty. The flow directions for pump A and pump B are given in the figure. Pump A can transfer whole water in tank A in 20 mins while pump B transfer whole water inside tank B in 100 min.

Pump A and Pump B are activated/deactivated in 10 minutes intervals.

∴ In this problem, we want to analyze the water levels in tank A tank B.

a-) find the differential equation set describing the system.

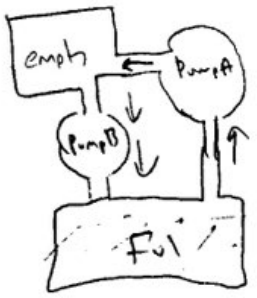
b-) write the state transition matrix

c-) show the characteristic equation of system.

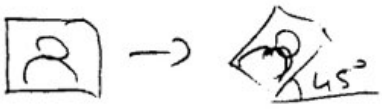
d-) determine the eigenvalues and eigen vectors

e-) what will be the water levels in both tanks after 6 hour?

f-) find the steady state water levels.



Q4- Consider a linear transformation that rotates an image 45 degrees. Determine the transformation matrix.



Solution=>

according the clock direction

$$\Rightarrow \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

$\theta = 45$ degrees.

$$\begin{bmatrix} \cos 45^\circ & \sin 45^\circ \\ -\sin 45^\circ & \cos 45^\circ \end{bmatrix}$$

$\cos 45^\circ \Rightarrow \frac{1}{\sqrt{2}}$

$\sin 45^\circ \Rightarrow \frac{1}{\sqrt{2}}$

So, the transformation matrix will be;

$$A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$